

Week-4

4.3.7 Fast Ethernet

5.2

What leads to fast Ethernet?

- ▶ To pump up the speed, various industry groups proposed two new ring based optical LANs.
- ▶ One was called FDDI and the other was called fiber channel.
- ▶ Both of them failed for desktop, management was complicated, complex chips and very expensive

Ethernet



Physical Media :-

- 10 Base5 - Thick Co-axial Cable with Bus Topology
- 10 Base2 - Thin Co-axial Cable with Bus Topology
- 10 BaseT - UTP Cat 3/5 with Tree Topology
- 10 BaseFL - Multimode/Singlemode Fiber with Tree Topology

Maximum Segment Length

- 10 Base5 - 500 m with at most 4 repeaters (Use Bridge to extend the network)
- 10 Base2 - 185 m with at most 4 repeaters (Use Bridge to extend the network)
- 10 BaseT - 100 m with at most 4 hubs (Use Switch to extend the network)

Fast Ethernet

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- ❏ 100 Mbps bandwidth
 - ❏ Uses same CSMA/CD media access protocol and packet format as in Ethernet.
 - ❏ 100BaseTX (UTP) and 100BaseFX (Fiber) standards
 - ❏ Physical media :-
 - ❏ 100 BaseTX - UTP Cat 5e
 - ❏ 100 BaseFX - Multimode / Singlemode Fiber
 - ❏ Full Duplex/Half Duplex operations.

Fast Ethernet

-  Provision for Auto-Negotiation of media speed:
10 Mbps or 100Mbps (popularly available for copper media only).
-  Maximum Segment Length
 -  100 Base TX - 100 m
 -  100 Base FX - 2 Km (Multimode Fiber)
 -  100 Base FX - 20 km (Singlemode Fiber)

Gigabit Ethernet



- 1 Gbps bandwidth.
- Uses same CSMA/CD media access protocol as in Ethernet and is backward compatible (10/100/100 modules are available).
- 1000BaseT (UTP), 1000BaseSX (Multimode Fiber) and 1000BaseLX (Multimode/Singlemode Fiber) standards.
- Maximum Segment Length
 - 1000 Base T - 100m (Cat 5e/6)
 - 1000 Base SX - 275 m (Multimode Fiber)
 - 1000 Base LX - 512 m (Multimode Fiber)
 - 1000 Base LX - 20 Km (Singlemode Fiber)
 - 1000 Base LH - 80 Km (Singlemode Fiber)

Routing Algorithm

- ▶ Part of N/L
- ▶ Decides 2 things:
 - ▶ Which i/p pkt to transmit?
 - ▶ Where/Which o/p line to select?
- ▶ Session routing: for the whole session no need to change the output line selected by 1st packet
- ▶ In general routers have 2 phase:
 - ▶ Forwarding
 - ▶ Table updating

Non-adaptive & adaptive algorithm

- ▶ Routing algorithm can be grouped into:

- ▶ **Non-adaptive algorithms** (fixed traffic light)

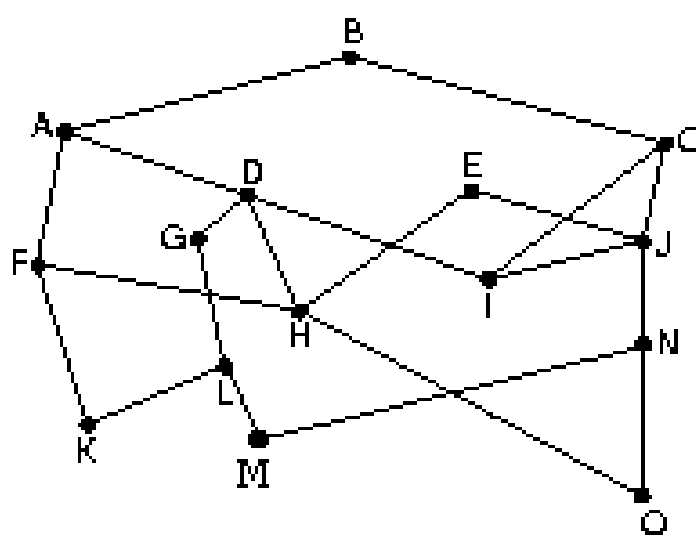
- ▶ do not base their routing decisions on measurements or estimates of the current traffic and topology.
 - ▶ Instead, the choice of the route to use to get from I to J (for all I and J) is computed in advance, off-line, and downloaded to the routers when the network is booted.
 - ▶ This procedure is sometimes called static routing.

- **Adaptive algorithms**

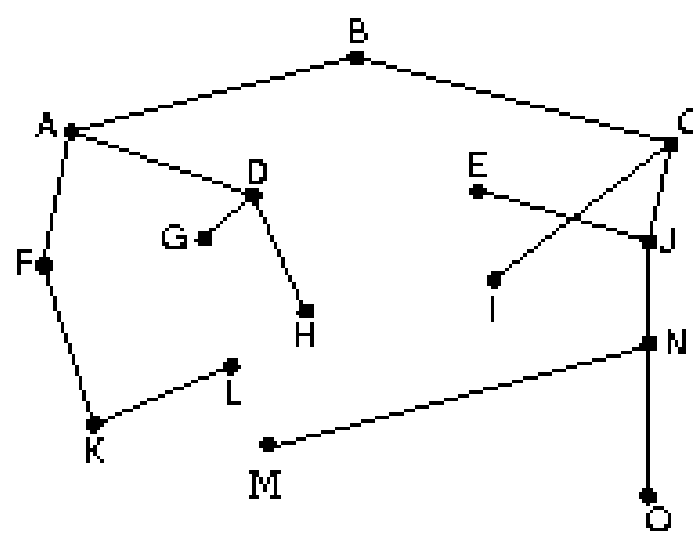
- ▶ change their routing decisions to reflect changes in the topology, and usually the traffic as well.
- ▶ Adaptive algorithms differ in where they get their
 - ▶ information (e.g., locally, from adjacent routers, or from all routers),
 - ▶ when they change the routes (e.g., every ΔT sec, when the load changes or when the topology changes), and
 - ▶ What metric is used for optimization (e.g., distance, number of hops, or estimated transit time).

The Optimality Principle

- ▶ the **optimality principle** and the sink tree provide a benchmark against which other routing algorithms can be measured.
 - ▶ a **general** statement about optimal routes without regard to network topology or traffic
 - ▶ It **states** that if router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route.



(a)



(b)

(a) A subnet. (b) A sink tree for router *B*.

- Where the **distance metric** is the number of hops.
- Sink tree is not necessarily unique; other trees with the same path lengths may exist.
- The goal of all routing algorithms is to discover and use the sink trees for all routers.

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- ▶ Design Goals of Routing Algorithms
 - ▶ Algorithm Types
 - ▶ Routing Metrics
 - ▶ Dynamic Routing Algorithms

Algorithm types

- Static Vs Dynamic
- Single Path Vs Multiple Path
- Host Intelligent Vs Router Intelligent
- Intradomain Vs Interdomain

Routing Metrics

- Path Length
- Reliability
- Delay
- Bandwidth
- Load
- Communication cost

Static Algorithms

- Shortest Path Algorithm
- Flooding
- Flow based Routing

Dynamic Algorithms

- Distance Vector Routing
- Link State Routing
- Hierarchical Routing

Shortest Path Routing (a non-adaptive routing algorithm)

- ▶ **Given** a network topology and a **set of weights** describing the **cost** to send data across each link in the network
- ▶ Find the shortest path from a specified source to all other destinations in the network.
- ▶ Shortest path algorithm first developed by E. W. Dijkstra

Mark the source node as **permanent**.

Designate the source node as the **working node**.

Set the **tentative** distance to all other nodes to **infinity**.

While some nodes are not marked permanent

Compute the tentative distance from the source to all nodes **adjacent** to the working node. If this is shorter than the current tentative distance replace the tentative distance of the destination and record the **label of the working node** there.

Examine ALL tentatively labeled nodes in the graph. **Select the node with the smallest value** and make it the new working node.

Example of Shortest Path Routing

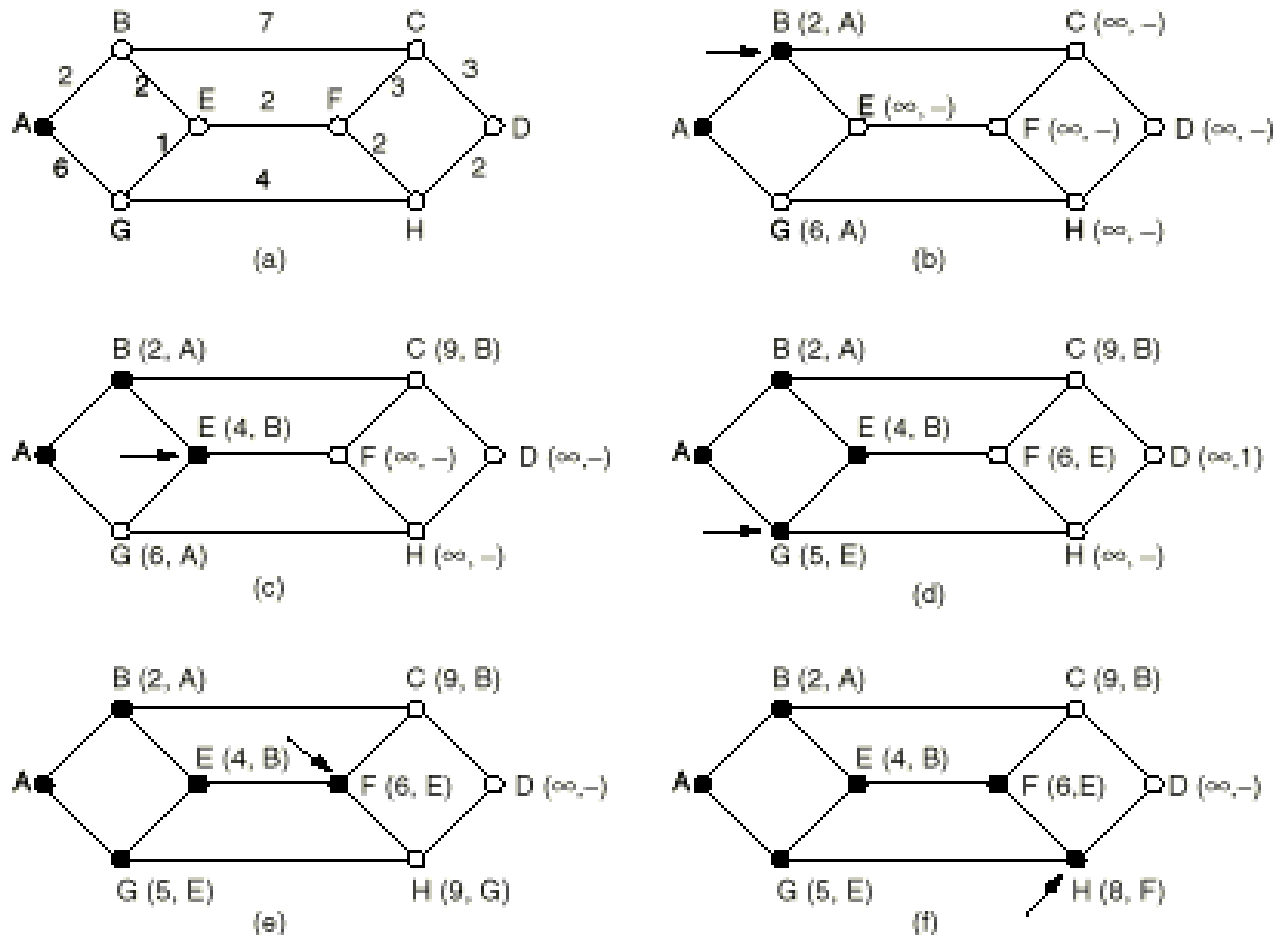


Fig. 5-6. The first five steps used in computing the shortest path from A to D. The arrows indicate the working node.

Flooding

(a nonadaptive routing algorithm)

- ▶ Brute force routing
 - ▶ Every incoming packet is sent on every outgoing line
 - ▶ Always finds the shortest path quickly
 - ▶ Also finds many long paths
 - ▶ Time to live is set to size of subnet
- ▶ Selective Flooding
 - ▶ Flood only in the direction of the destination
- ▶ Practical in a few settings
 - ▶ Military Applications
 - ▶ Distributed Databases
 - ▶ Metric for comparison

Distance Vector Routing

(an adaptive routing algorithm)

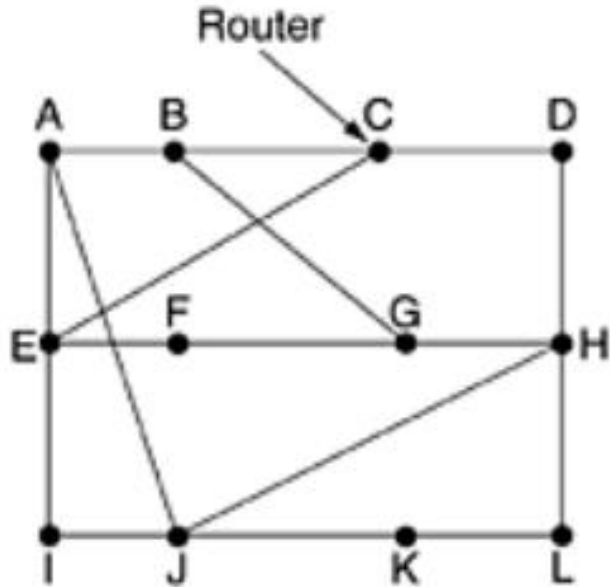
- ▶ Bellman-Ford Routing
- ▶ Ford Fulkerson Algorithm
- ▶ Original ARPANET routing algorithm
- ▶ Previously used on Internet (RIP)
- ▶ Early version of DecNet and Novell's IPX
- ▶ AppleTalk and Cisco routers use improved versions of this algorithm

Distance Vector Routing (an adaptive routing algorithm)

- ▶ Neighboring routers periodically exchange information from their routing tables.
- ▶ Routers replace routes in their own routing tables anytime that neighbors have found better routes.
- ▶ Information provided from neighbors
 - ▶ Outgoing line used for destination
 - ▶ Estimate of time or distance
 - ▶ can be number of hops, time delay, packet queue length, etc.

Distance Vector Routing

(an adaptive routing algorithm)



To	A	I	H	K	New estimated delay from J	
					↓	Line
A	0	24	20	21		
B	12	36	31	28		
C	25	18	19	36		
D	40	27	8	24		
E	14	7	30	22		
F	23	20	19	40		
G	18	31	6	31		
H	17	20	0	19		
I	21	0	14	22		
J	9	11	7	10		
K	24	22	22	0		
L	29	33	9	9		

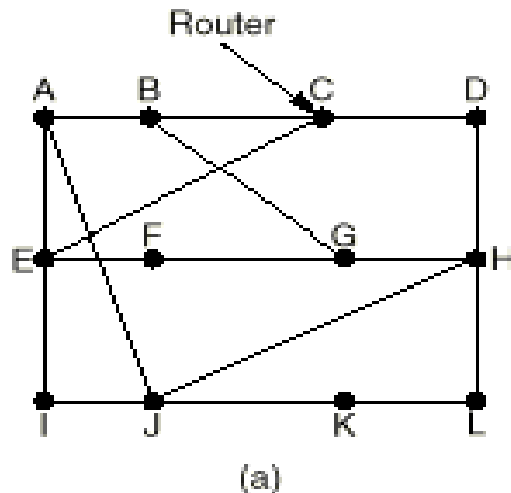
JA delay is 8 JI delay is 10 JH delay is 12 JK delay is 6

Vectors received from J's four neighbors

New routing table for J

Distance Vector Routing

(an adaptive routing algorithm)



New estimated delay from J

To	A	I	H	K	New estimated delay from J	Line
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	—
K	24	22	22	0	6	K
L	29	33	9	9	15	K

To	A	I	H	K
A	0	24	20	21
B	12	36	31	28
C	25	18	19	36
D	40	27	8	24
E	14	7	30	22
F	23	20	19	40
G	18	31	6	31
H	17	20	0	19
I	21	0	14	22
J	9	11	7	10
K	24	22	22	0
L	29	33	9	9

JA delay is 8 JI delay is 10 JH delay is 12 JK delay is 6

Vectors received from J's four neighbors

New routing table for J

(b)

Fig. 5-10. (a) A subnet. (b) Input from *A*, *I*, *H*, *K*, and the new routing table for *J*.

The Count to Infinity Problem

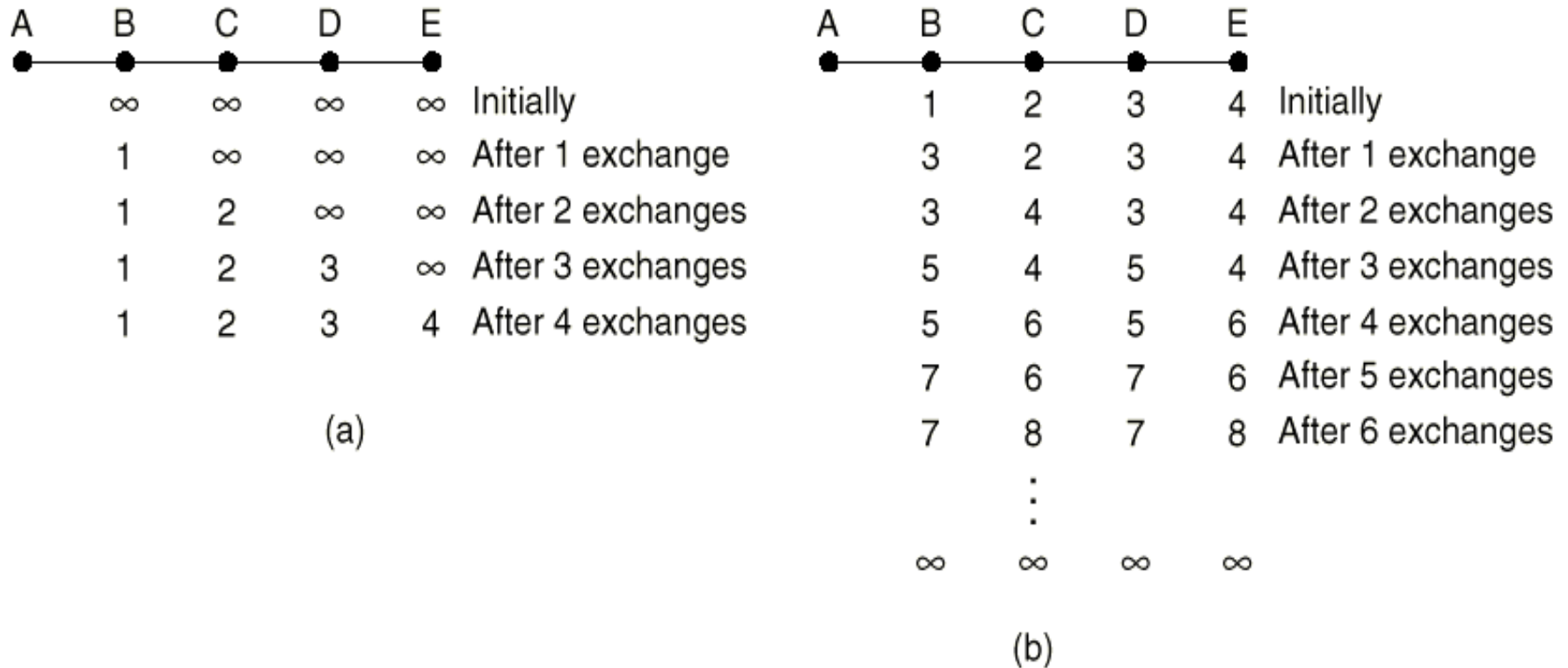


Fig. 5-11. The count-to-infinity problem.

Link State Routing

(an adaptive routing algorithm)

► Five Steps

- 1.) Discover your neighbors and learn their addresses.
- 2.) Measure the cost (delay) to each neighbor.
- 3.) Construct a packet containing all this information
- 4.) Send this packet to all other routers.
- 5.) Compute the shortest path to every other router.

1.) Discovering Your Neighbors

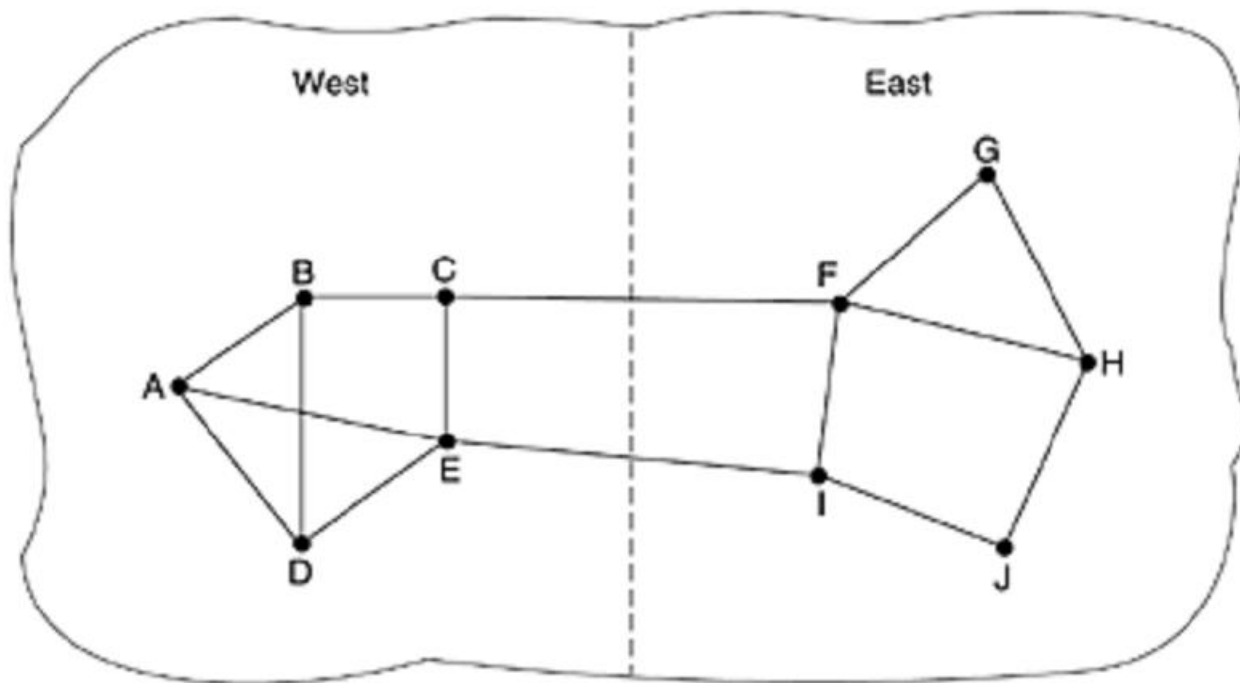
Send “Hello” packet on each point-to-point line. Destination node replies with its address.

2.) Measuring Line Cost

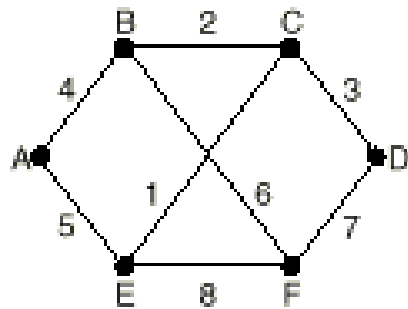
- ▶ Send an “ECHO” packet over the line.
- ▶ Destination is required to respond to “ECHO” packet immediately.
- ▶ Measure the time required for this operation.
- ▶ Question: Should we measure just the time it takes to transmit the packet, or should we include the time that the packet waits in the queue?

Argument 2:

- ▶ We should include the time that the packet spends in the queue, as this provides a more accurate picture of the real delays.
- ▶ We should only include the transmission times, otherwise the network is likely to oscillate between preferred paths.



Build Link State Packets



(9)

		Link	State			Packets			
A		B	C	D		E	F		
Seq.		Seq.	Seq.	Seq.		Seq.	Seq.		
Age		Age	Age	Age		Age	Age		
B	4	A	4	B	2	A	5	B	6
E	5	C	2	D	3	C	1	D	7
		F	6	E	1	F	8	E	8

(b)

Fig. 5-15. (a) A subnet. (b) The link state packets for this subnet.

Building the link state packets is easy.
when to build? 1) periodically, and
2) when some significant event occurs,

Distributing the Link State Packets

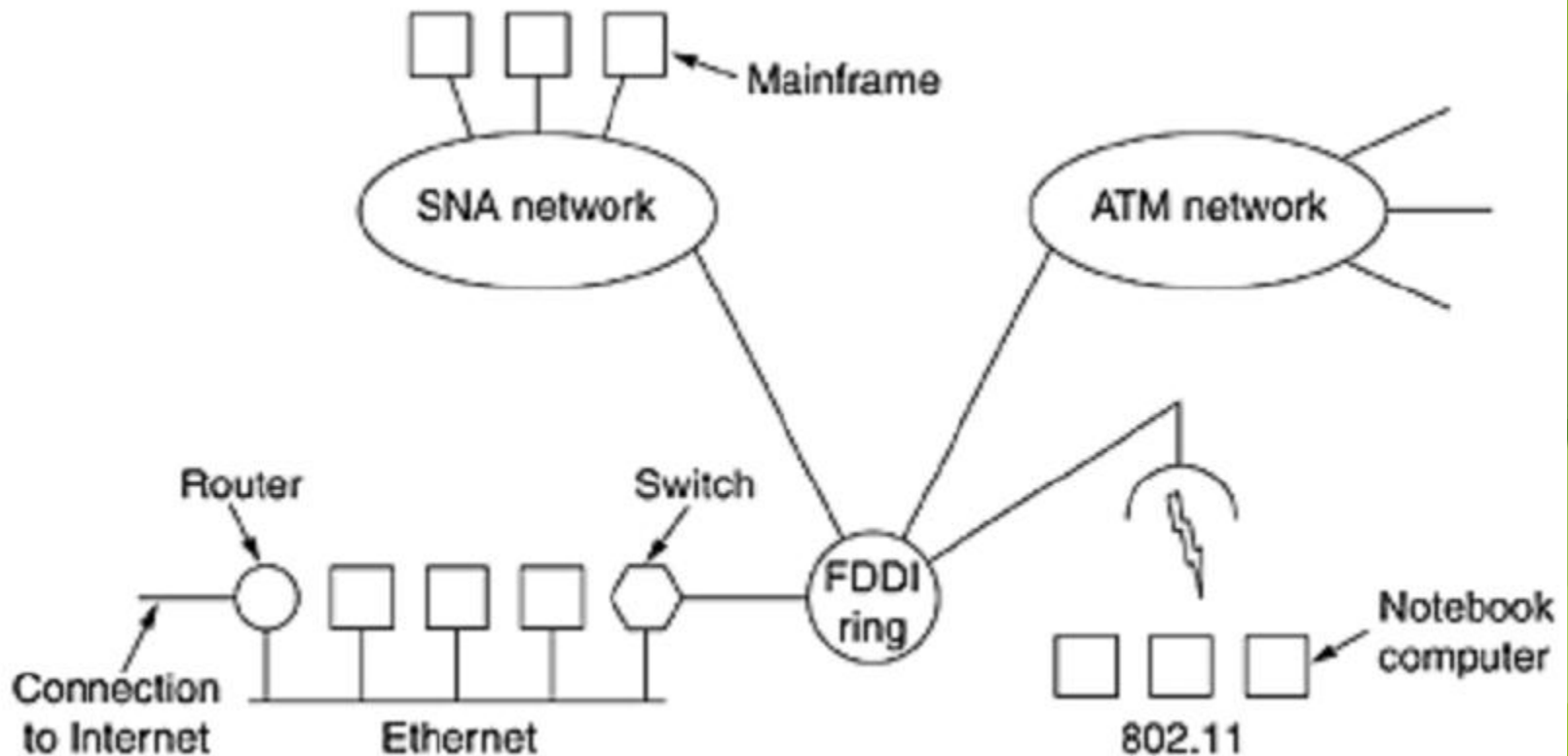
- ▶ Use selective flooding
- ▶ **Sequence numbers** prevent duplicate packets from being propagated
- ▶ Lower sequence numbers are rejected as obsolete

Computing the New Routes

Dijkstra's Shortest Path algorithm is used to determine the shortest path to each destination.

Internetworking

- More networks are connected to form internet



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- ▶ The **purpose** of interconnecting all these networks is to allow users on any of them to **Communicate** with users on all the other ones and also to allow users on any of them to access data on any of them.

How Networks Differ

Item	Some Possibilities
Service offered	Connection oriented versus connectionless
Protocols	IP, IPX, SNA, ATM, MPLS, AppleTalk, etc.
Addressing	Flat (802) versus hierarchical (IP)
Multicasting	Present or absent (also broadcasting)
Packet size	Every network has its own maximum
Quality of service	Present or absent; many different kinds
Error handling	Reliable, ordered, and unordered delivery
Flow control	Sliding window, rate control, other, or none
Congestion control	Leaky bucket, token bucket, RED, choke packets, etc.
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, by packet, by byte, or not at all

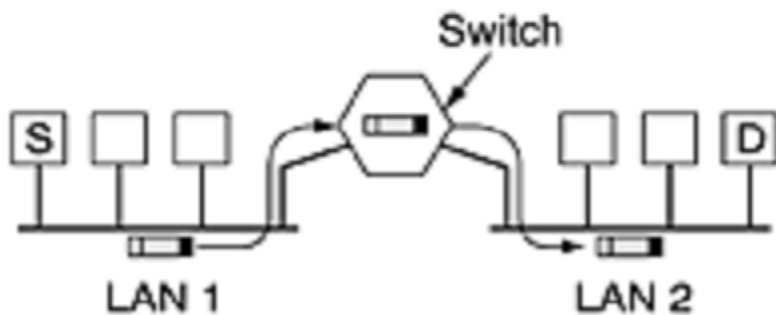
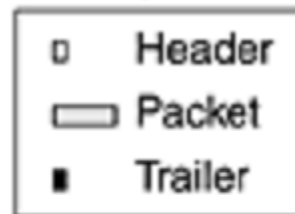
How Networks Can Be Connected?

- ▶ Repeaters or Hubs (which layer?)
- ▶ Bridges and Switches. (?)
 - ▶ Can translate protocol from Ethernet to FDDI or to 802.11.
(minor protocol translation)
- ▶ Routers (?)
 - ▶ A router that can handle multiple protocols is called a multiprotocol router.

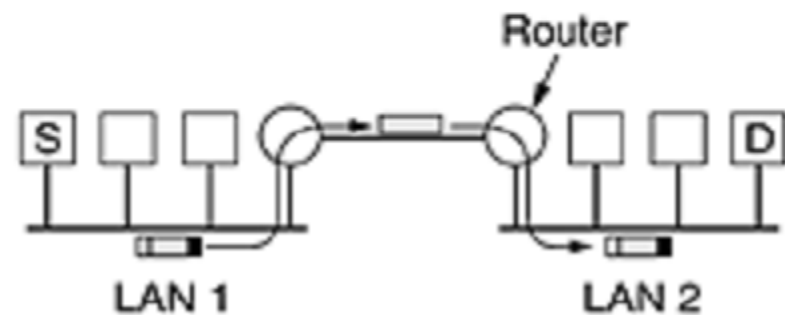
Switch vs Router

- ▶ Switch only checks MAC address, but
- ▶ Router checks the IP address.

Legend



(a)



(b)

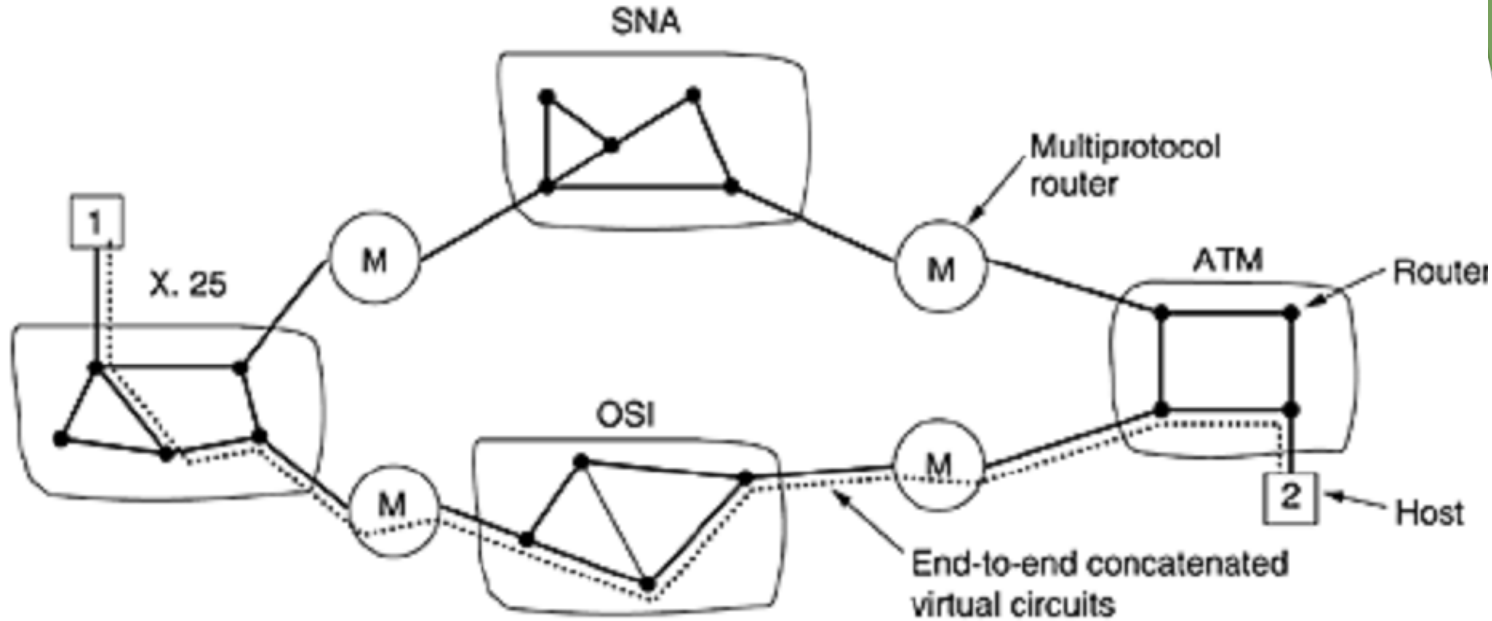


Figure 5-45. Internetworking using concatenated virtual circuits.

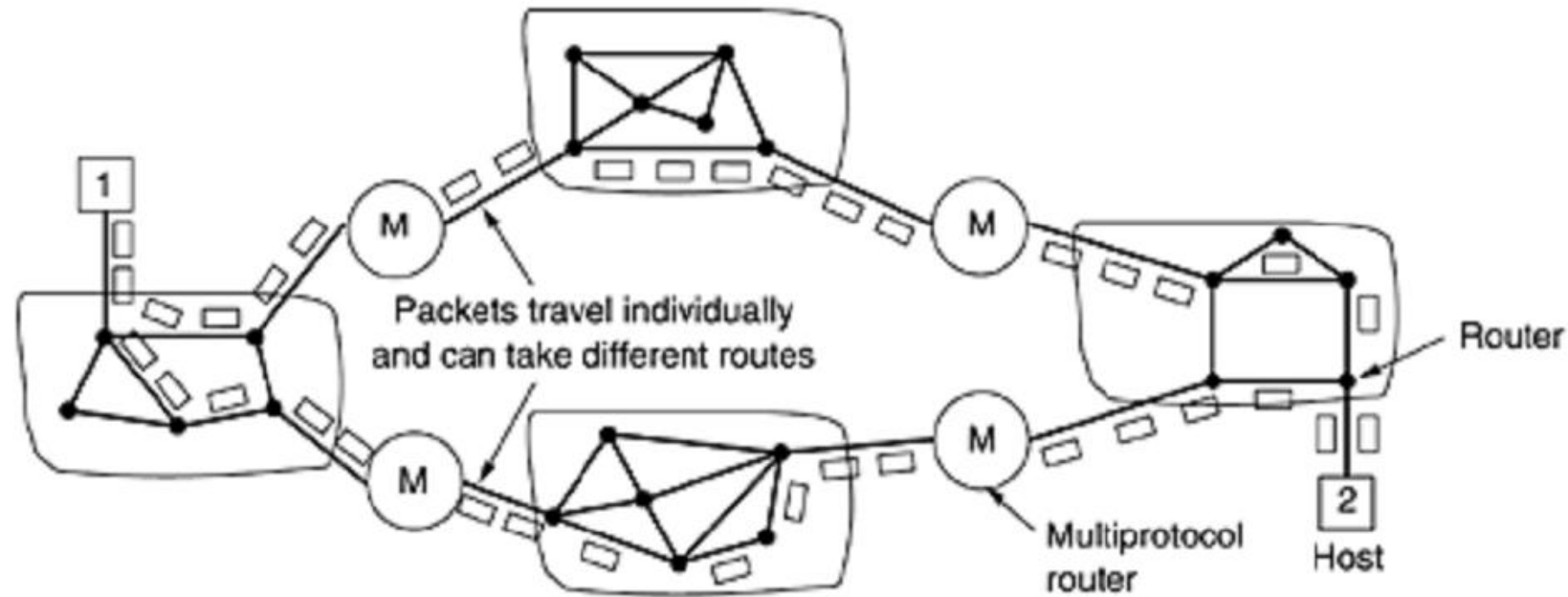


Figure 5-46. A connectionless internet.

Tunneling

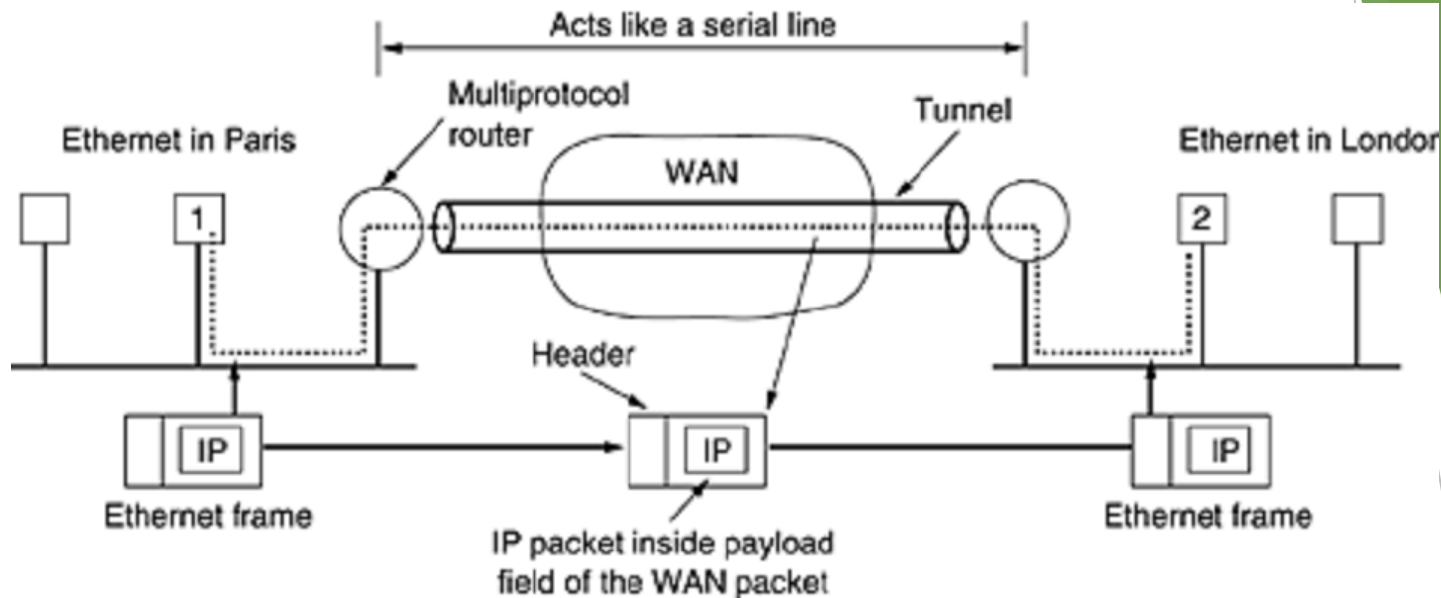


Figure 5-47. Tunneling a packet from Paris to London.

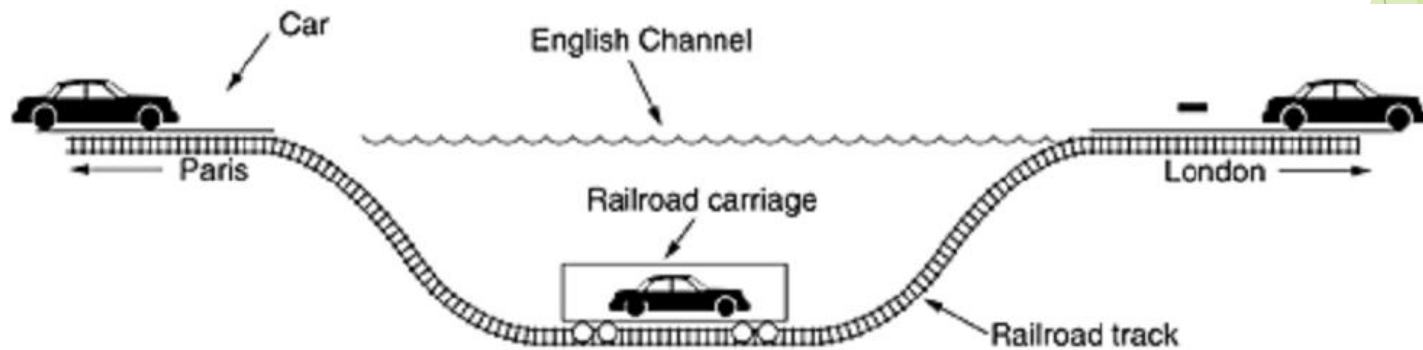
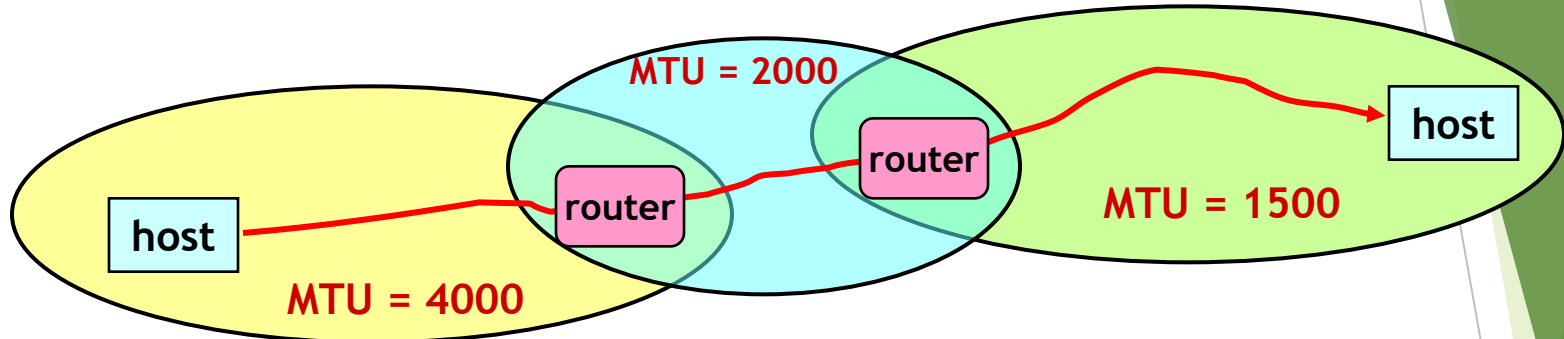


Figure 5-48. Tunneling a car from France to England.

Fragmentation. Why?

1. Hardware (e.g., the size of an Ethernet frame).
2. Operating system (e.g., all buffers are 512 bytes).
3. Protocols (e.g., the number of bits in the packet length field).
4. Compliance with some (inter)national standard.
5. Desire to reduce error-induced retransmissions to some level.
6. Desire to prevent one packet from occupying the channel too long.

IP Fragmentation



- ▶ Every Network has Own Maximum Transmission Unit (MTU)
 - ▶ Largest IP datagram it can carry within its own packet frame
 - ▶ E.g., Ethernet is 1500 bytes
 - ▶ Don't know MTUs of all intermediate networks in advance
- ▶ IP Solution
 - ▶ When hit network with small MTU, fragment packets
 - ▶ Might get further fragmentation as proceed farther

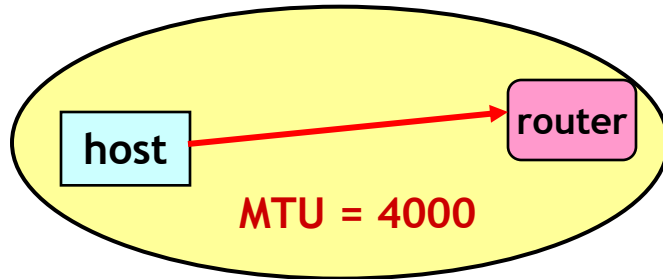
Fragmentation Related Fields

- ▶ Length
 - ▶ Length of IP fragment
- ▶ Identification
 - ▶ To match up with other fragments
- ▶ Fragment offset
 - ▶ Where this fragment lies in entire IP datagram
- ▶ Flags
 - ▶ “More fragments” flag
 - ▶ “Don’t fragment” flag

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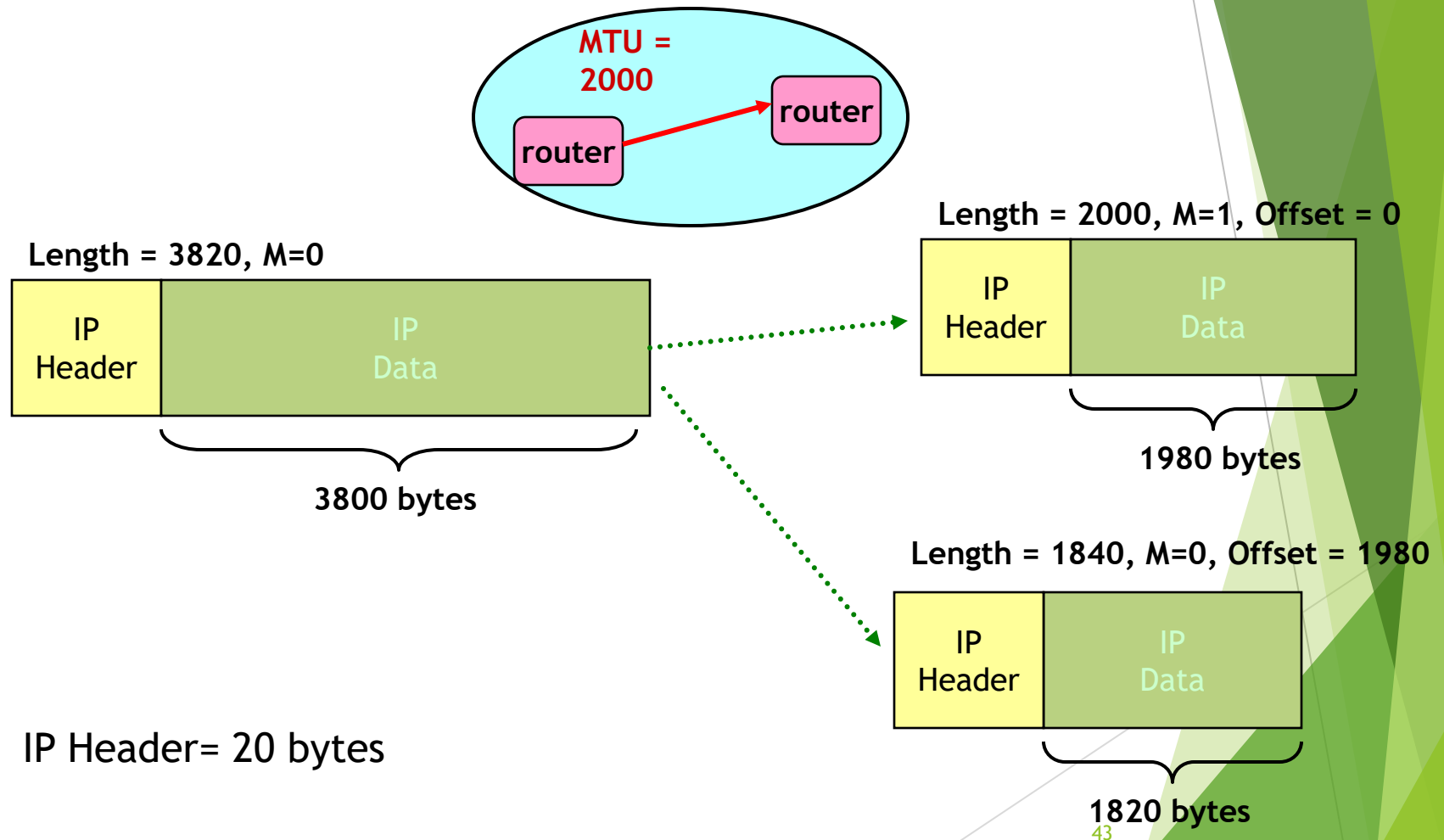
IP Fragmentation Example #1



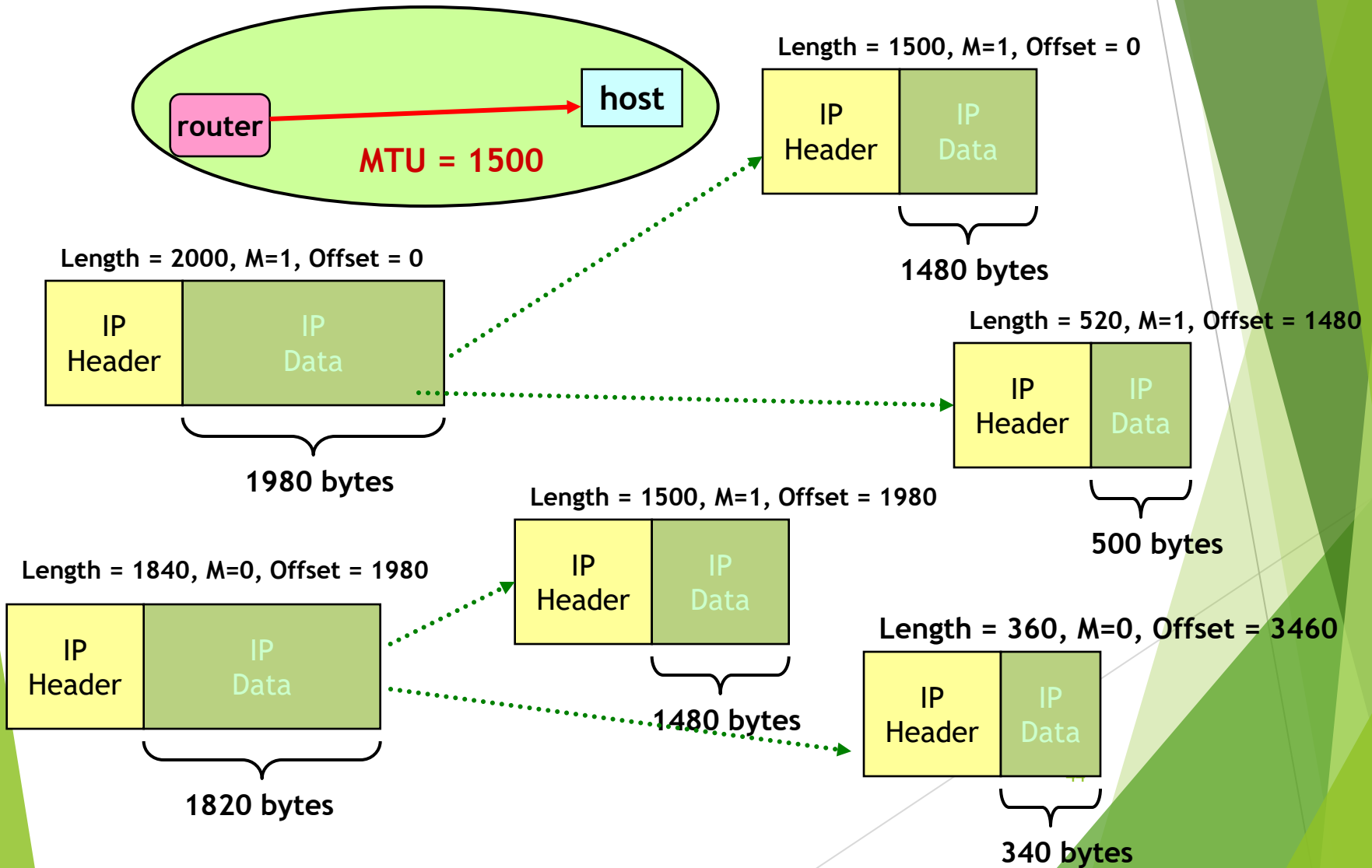
Length = 3820, M=0



IP Fragmentation Example #2

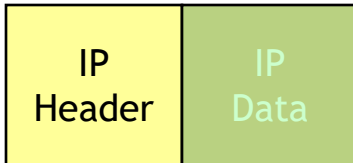


IP Fragmentation Example #3



IP Reassembly

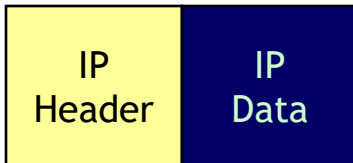
Length = 1500, M=1, Offset = 0



Length = 520, M=1, Offset = 1480



Length = 1500, M=1, Offset = 1980



Length = 360, M=0, Offset = 3460



- ▶ Fragments might arrive out-of-order
 - ▶ Don't know how much memory required until receive final fragment
- ▶ Some fragments may never arrive
 - ▶ After a while, give up entire process

